1) Najděte $\vec{u} = \overrightarrow{AB}$ nakreslete 3 umístění vektoru $\vec{u}$ a najděte X tak, aby $\overrightarrow{CX} = \vec{u}$

$A = [1,3]$ $B = [4,1]$ $C = [-3,-2]$

$\vec{u} = \overrightarrow{AB} = (b_x - a_x, b_y - a_y)$
$= (3, -2)$

umístění $\vec{u}$ do $D = [-2,2]$:
$D + \vec{u} = \begin{pmatrix}-2\\2\end{pmatrix} + \begin{pmatrix}3\\-2\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$

$X = [?,?]$ tak, aby $\overrightarrow{CX} = \vec{u}$

$X = C + \vec{u} \iff \vec{u} = \text{"}X - C\text{"} = \overrightarrow{CX}$

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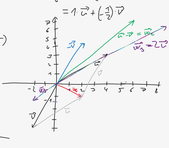
2) $\vec{u} = (4,3)$ $\vec{v} = (-2,-4)$

$\vec{w} = (2,-1)$
$\vec{w_1} = (6, ?)$
$\vec{w_2} = (8,6)$
$\vec{w_3} = (4,3) + (1,2) = (5,5)$

opačný k $\vec{v}$: $-\vec{v} = (2,4)$

$\vec{w_4} = -\frac{1}{2}\vec{u} = \left(-2, -\frac{3}{2}\right)$

$\vec{w_5} = \vec{u} + \vec{v}$
$\vec{w_6} = \vec{u} - \vec{v}$
$\vec{w_7} = 2\vec{u}$
$\vec{w_8} = 2\vec{u} - \vec{v}$
$= 2\vec{u} + \left(-\frac{1}{2}\right)\vec{v}$



Velikost vektoru

o $\vec{u}$: $|\overrightarrow{AB}| = \sqrt{\sum_{i=1}^{n} (b_i - a_i)^2}$ $\mathbb{R}^2$: $|\overrightarrow{AB}| = \sqrt{(b_x - a_x)^2 + (b_y - a_y)^2}$

$\overrightarrow{AB} = (b_x - a_x, b_y - a_y)$
$\vec{u} = (u_1, u_2)$

$|\vec{u}| = \sqrt{u_1^2 + u_2^2}$

$\mathbb{R}^n$: $|\vec{u}| = \sqrt{\sum_{i=1}^{n} u_i^2}$

Př: $\vec{u} = (1,1)$ $|\vec{u}| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\vec{u} = (2, 0, 3)$ Najděte rovnoběžný vektor s velikostí 1

$|\vec{u}| = \sqrt{2^2 + 0^2 + 3^2} = \sqrt{13}$ $\vec{u} = \nearrow$

$\vec{v} = \alpha \vec{u}$ $|\vec{v}| = 1$ $\alpha \in \mathbb{R}$ "normalizace"

$\vec{v} = \frac{1}{\sqrt{13}}\vec{u} = \left(\frac{2}{\sqrt{13}}, 0, \frac{3}{\sqrt{13}}\right)$

$|\vec{v}| = \sqrt{\left(\frac{2}{\sqrt{13}}\right)^2 + 0^2 + \left(\frac{3}{\sqrt{13}}\right)^2} = \sqrt{\frac{4}{13} + \frac{9}{13}} = \sqrt{\frac{13}{13}} = 1$

Skalární součin

$\vec{u} \cdot \vec{v} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + \dots + u_n v_n$

$|\vec{u}| = \sqrt{\sum_{i=1}^{n} u_i^2} = \sqrt{\sum_{i=1}^{n} u_i \cdot u_i} = \sqrt{\vec{u} \cdot \vec{u}} \iff \vec{u} \cdot \vec{u} = |\vec{u}|^2$

$\vec{u} = (-3, 1, -2) \longrightarrow |\vec{u}|^2 = 9 + 1 + 4 = 14$
$\longrightarrow |\vec{u}| = \sqrt{14}$

Skalární součin a úhel

$\boxed{\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos\varphi}$

$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \left(\frac{\vec{u}}{|\vec{u}|}\right) \cdot \left(\frac{\vec{v}}{|\vec{v}|}\right)$

$|\ |$ ... $|\cdot| = 1$

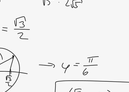
Př: Najděte úhel, který svírají vektory $\vec{u}$ a $\vec{v}$

$\vec{u} = (1,1)$ $\vec{v} = (2, 0)$

$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \frac{1 \cdot 2 + 1 \cdot 0}{2\sqrt{2}} = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$

$|\vec{u}| = \sqrt{1^2 + 1^2} = \sqrt{2}$
$|\vec{v}| = \sqrt{2^2 + 0^2} = 2$

$\cos\varphi = \frac{\sqrt{2}}{2}$



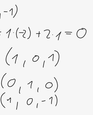
$\varphi_1 = \frac{\pi}{4}$
$\varphi_2 = \frac{7}{4}\pi$

$|\ |$ ... velikost

- $\vec{u} = (-1, 2)$ $\vec{v} = (-\sqrt{3} + 2, 1 + 2\sqrt{3})$
  $\varphi = ?$
  $|\vec{u}| = \sqrt{1^2 + 2^2} = \sqrt{5}$
  $|\vec{v}| = \sqrt{(-\sqrt{3}+2)^2 + (1+2\sqrt{3})^2} = \sqrt{3 - 4\sqrt{3} + 4 + 1 + 4\sqrt{3} + 4 \cdot 3} = \sqrt{20} = \sqrt{5 \cdot 4} = \sqrt{5}\sqrt{4} = 2\sqrt{5}$

$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} = \frac{-1(-\sqrt{3}+2) + 2(1+2\sqrt{3})}{\sqrt{5} \cdot 2\sqrt{5}} = \frac{\sqrt{3} - 2 + 2 + 4\sqrt{3}}{2 \cdot 5} = \frac{5\sqrt{3}}{2 \cdot 5} = \frac{\sqrt{3}}{2}$

$\cos\varphi = \frac{\sqrt{3}}{2}$



$\longrightarrow \varphi = \frac{\pi}{6}$

$\cos\varphi = \frac{\sqrt{3}}{2}$ $\varphi = \arccos\left(\frac{\sqrt{3}}{2}\right)$

$\begin{pmatrix}\sqrt{3} & 1\\ -1 & \sqrt{3}\end{pmatrix} \cdot \vec{u}$

$\vec{u} \cdot \vec{v} = |\vec{u}| \cdot |\vec{v}| \cdot \cos\varphi$

$\varphi = \frac{\pi}{2} \longrightarrow \cos\frac{\pi}{2} = 0 \longrightarrow \boxed{\vec{u} \cdot \vec{v} = 0}$

$\boxed{\vec{u} \perp \vec{v} \implies \vec{u} \cdot \vec{v} = 0}$ Kolmé vektory "ortogonální"

pokud navíc $|\vec{u}| = 1$ $|\vec{v}| = 1$
$\longrightarrow$ "ortonormální"

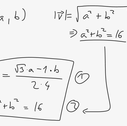
- Rozhodněte zda $\vec{u} = (1, 2, -1) \perp \vec{v} = (0, -3, 2)$

$\vec{u} \cdot \vec{v} = 0 - 6 - 2 = -8$ NE

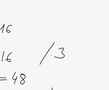
- Najděte $\vec{v}$ tak, aby $\vec{u} \cdot \vec{v} = 0$ $(\vec{u} \perp \vec{v})$

$\vec{u} = (1, 2)$
$\vec{v} = (-2, 1)$
nebo $(2, -1)$

$\vec{u} \cdot \vec{v} = 1(-2) + 2 \cdot 1 = 0$



- $\mathbb{R}^3$: $\vec{u} = (1, 0, 1)$
  $\vec{v_1} = (0, 1, 0)$
  $\vec{v_2} = (1, 0, -1)$



$\vec{w} = a \cdot \vec{v_1} + b \cdot \vec{v_2}$ $a, b \in \mathbb{R}$
$\vec{w} \perp \vec{u}$

- Je dán $\vec{u}$. Najděte $\vec{v}$ tak, aby svíraly úhel $\varphi = 60°$ a $|\vec{v}| = 4$

$\vec{u} = (\sqrt{3}, -1)$

$\cos 60° = \frac{1}{2}$ $\vec{v} = (a, b)$ $|\vec{v}| = \sqrt{a^2 + b^2} = 4$
$\Rightarrow a^2 + b^2 = 16$

$|\vec{u}| = \sqrt{3 + 1} = 2$

$\cos\varphi = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| \cdot |\vec{v}|} \longrightarrow \boxed{\begin{aligned} \frac{1}{2} &= \frac{\sqrt{3} \cdot a - 1 \cdot b}{2 \cdot 4} \quad (1) \\ a^2 + b^2 &= 16 \quad (2) \end{aligned}}$

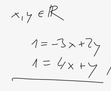
(1) $\sqrt{3}a - b = 4$
$a = \frac{1}{\sqrt{3}}(b + 4)$

(2) $\left(\frac{1}{\sqrt{3}}(b+4)\right)^2 + b^2 = 16$

$\frac{1}{3}(b^2 + 8b + 16) + b^2 = 16 \quad / 3$
$b^2 + 8b + 16 + 3b^2 = 48$
$4b^2 + 8b - 32 = 0 \quad / : 4$
$b^2 + 2b - 8 = 0$
$(b + 4)(b - 2) = 0 \longrightarrow b_1 = -4$, $b_2 = 2$

(1): $a = \frac{1}{\sqrt{3}}(b + 4) \Rightarrow a_1 = \frac{1}{\sqrt{3}}(-4 + 4) = 0$
$a_2 = \frac{1}{\sqrt{3}}(2 + 4) = \frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\boxed{\begin{aligned} \vec{v_1} &= (0, -4) \\ \vec{v_2} &= (2\sqrt{3}, 2) \end{aligned}}$



Vektor $\vec{z} = \begin{pmatrix}1\\1\end{pmatrix}$ napište jako lineární kombinaci vektorů $\vec{w} = \begin{pmatrix}-3\\4\end{pmatrix}$ $\vec{v} = \begin{pmatrix}2\\1\end{pmatrix}$

$\vec{z} = \lambda_1 \vec{w} + \lambda_2 \vec{v}$ $x, y \in \mathbb{R}$

$\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}-3x + 2y\\4x + y\end{pmatrix}$

$\left.\begin{aligned} 1 &= -3x + 2y \\ 1 &= 4x + y \quad /2 \end{aligned}\right\} \ominus$

$-1 = -11x$
$\boxed{x = \frac{1}{11}}$

$1 = -\frac{3}{11} + 2y$
$\boxed{y = \frac{7}{11}}$

$\frac{1}{11}\overset{\vec{w}}{\begin{pmatrix}-3\\4\end{pmatrix}} + \frac{7}{11}\overset{\vec{v}}{\begin{pmatrix}2\\1\end{pmatrix}} = \begin{pmatrix}\frac{11}{11}\\\frac{11}{11}\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix} = \vec{z}$